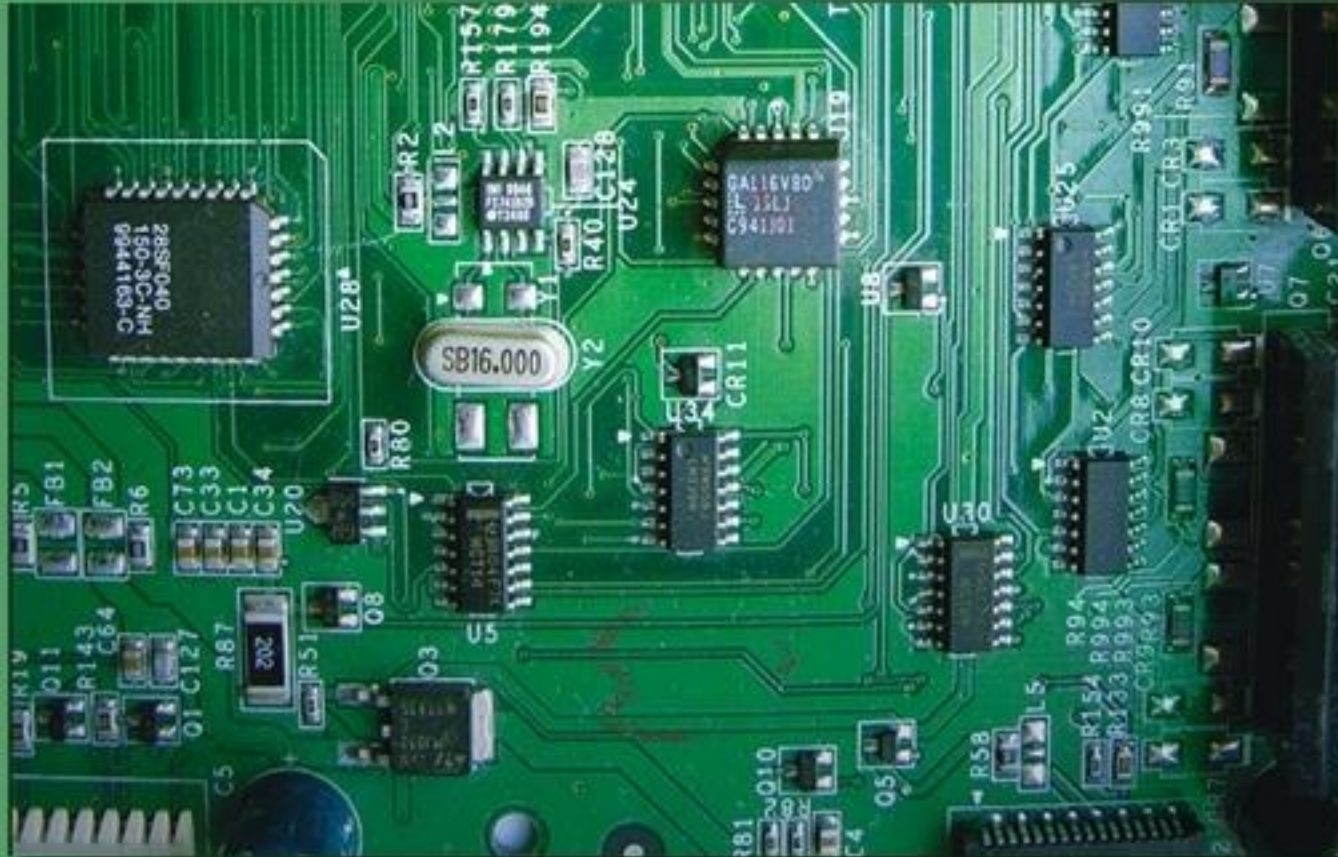


The Intel Microprocessors

8086/8088, 80186/80188, 80286, 80386, 80486 Pentium, Pentium Pro Processor, Pentium II, Pentium 4, and Core2 with 64-bit Extensions

Architecture, Programming, and Interfacing



EIGHTH EDITION

Barry B. Brey

PEARSON

Chapter 9: 8086/8088 Hardware Specifications

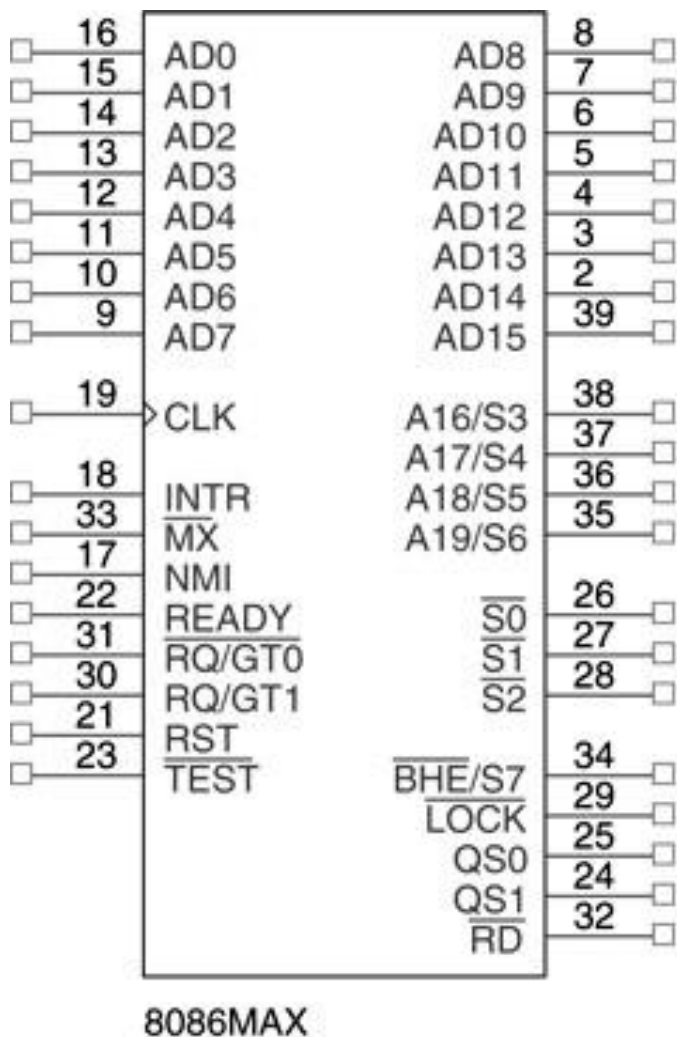
9-1 PIN-OUTS AND THE PIN FUNCTIONS

- In this section, we explain the function and the **multiple functions** of each of the microprocessor's pins.
- In addition, we discuss the **DC characteristics** to provide a basis for understanding the later sections on **buffering** and **latching**.

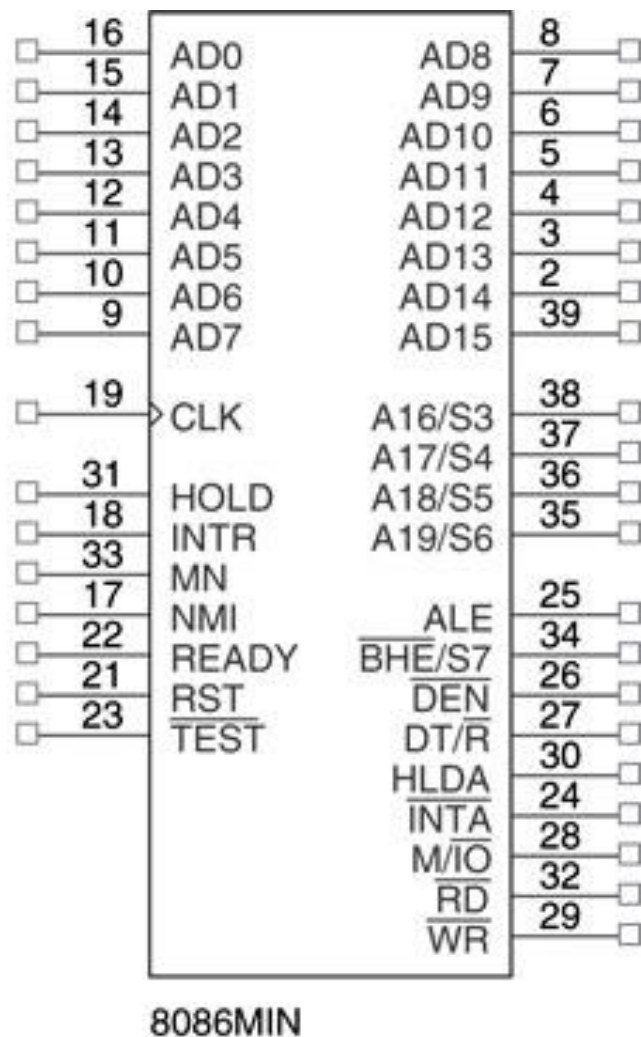
The Pin-Out

- Figure 9–1 illustrates pin-outs of 8086.
 - packaged in 40-pin **dual in-line** packages (**DIPs**)
- 8086 is a **16-bit** microprocessor with a 16-bit data bus;
 - 8086 has pin connections **AD₀–AD₁₅**
 - thus 8086 transfers 16-bit data more efficiently

Figure 9–1 (a) The pin-out of the 8086 in **maximum mode**; (b) the pin-out of the 8086 in **minimum mode**.



(a)



(b)

Pin Connections

$AD_7 - AD_0$

Pin Connections

$AD_{15} - AD_8$

- 8086 **address/data bus** lines compose upper multiplexed address/data bus on the 8086.
- These lines contain address bits $A_{15}-A_8$ whenever **ALE** is a logic 1, and data bus connections $D_{15}-D_8$ when **ALE** is a logic 0.

Pin Connections A_{19}/S_6 - A_{16}/S_3

- **Address/status bus** bits are multiplexed to provide address signals A_{19} – A_{16} and **status bits** S_6 – S_3 .
 - status bit S_6 is always logic 0,
 - bit S_5 indicates the condition of the IF flag bit
- S_4 and S_3 show which segment is accessed during the current bus cycle.

Pin Connections $\overline{RD}$

- When **read signal** is logic 0, the data bus is receptive to data from memory or I/O devices
 - pin floats high-impedance state during a hold acknowledge

Pin Connections INTR

- **Interrupt request** is used to request a hardware interrupt.
 - If INTR is held high when $IF = 1$, 8086/8088 enters an interrupt acknowledge cycle after the current instruction has completed execution

NMI

- The **non-maskable interrupt** input is similar to INTR.

Pin Connections RESET

- Causes the microprocessor to reset itself if held high a minimum of four clocking periods.
 - when 8086/8088 is reset, it executes instructions at memory location FFFF0H
 - also disables future interrupts by clearing IF flag

CLK

- The **clock** pin provides the **basic timing signal**.
 - must have a duty cycle of 33 % (high for one third of clocking period, low for two thirds) to provide proper internal timing

Pin Connections VCC

- This **power supply** input provides a **+5.0 V**, **±10 %** signal to the microprocessor.

GND

- The **ground** connection is the return for the power supply.
 - 8086/8088 microprocessors have two pins labeled GND—both must be connected to ground for proper operation

Pin Connections $\overline{MN}/\overline{MX}$

- **Minimum/maximum** mode pin selects either minimum or maximum mode operation.
 - if minimum mode selected, the $\overline{MN}/\overline{MX}$ pin must be connected directly to +5.0 V

BHE S_7

- The **bus high enable** pin is used in 8086 to enable the most-significant data bus bits (D_{15} – D_8) during a read or a write operation.
- The state of S_7 is always a logic 1.

Minimum Mode Pins

- Minimum mode operation is obtained by connecting the $\text{MN}/\overline{\text{MX}}$ pin directly to +5.0 V.
 - do not connect to +5.0 V through a pull-up register; it will not function correctly

$\text{IO}/\overline{\text{M}}$ or $\text{M}/\overline{\text{IO}}$

- The $\text{IO}/\overline{\text{M}}$ (8088) or $\text{M}/\overline{\text{IO}}$ (8086) pin **selects** memory or I/O.
 - indicates the address bus **contains either a memory address or an I/O port address.**

Minimum Mode Pins $\overline{WR}$

- **Write line** indicates 8086/8088 is outputting data to a memory or I/O device.
 - during the time $\overline{WR}$ is a logic 0, the data bus contains valid data for memory or I/O
 - high-impedance during a hold acknowledge

$\overline{INTA}$

- The **interrupt acknowledge** signal is a response to the **INTR** input pin.
 - normally used to gate the interrupt vector number onto the data bus in response to an interrupt

Minimum Mode Pins ALE

- **Address latch enable** shows the 8086/8088 address/data bus contains an address.
 - can be a memory address or an I/O port number
 - ALE signal **doesn't float** during hold acknowledge

DT/ $\overline{R}$

- The **data transmit/receive** signal shows that the microprocessor **data bus** is transmitting (**DT/ $\overline{R}$ = 1**) or receiving (**DT/ $\overline{R}$ = 0**) data.
 - used to enable external data bus buffers

Minimum Mode Pins DEN

- **Data bus enable** activates external data bus buffers.

9-3 BUS BUFFERING AND LATCHING

- Before 8086/8088 can be used with memory or I/O interfaces, their **multiplexed buses** must be **demultiplexed**.
- This section provides detail required to demultiplex the buses and illustrates how the **buses** are buffered for very large systems.

Demultiplexing the Buses

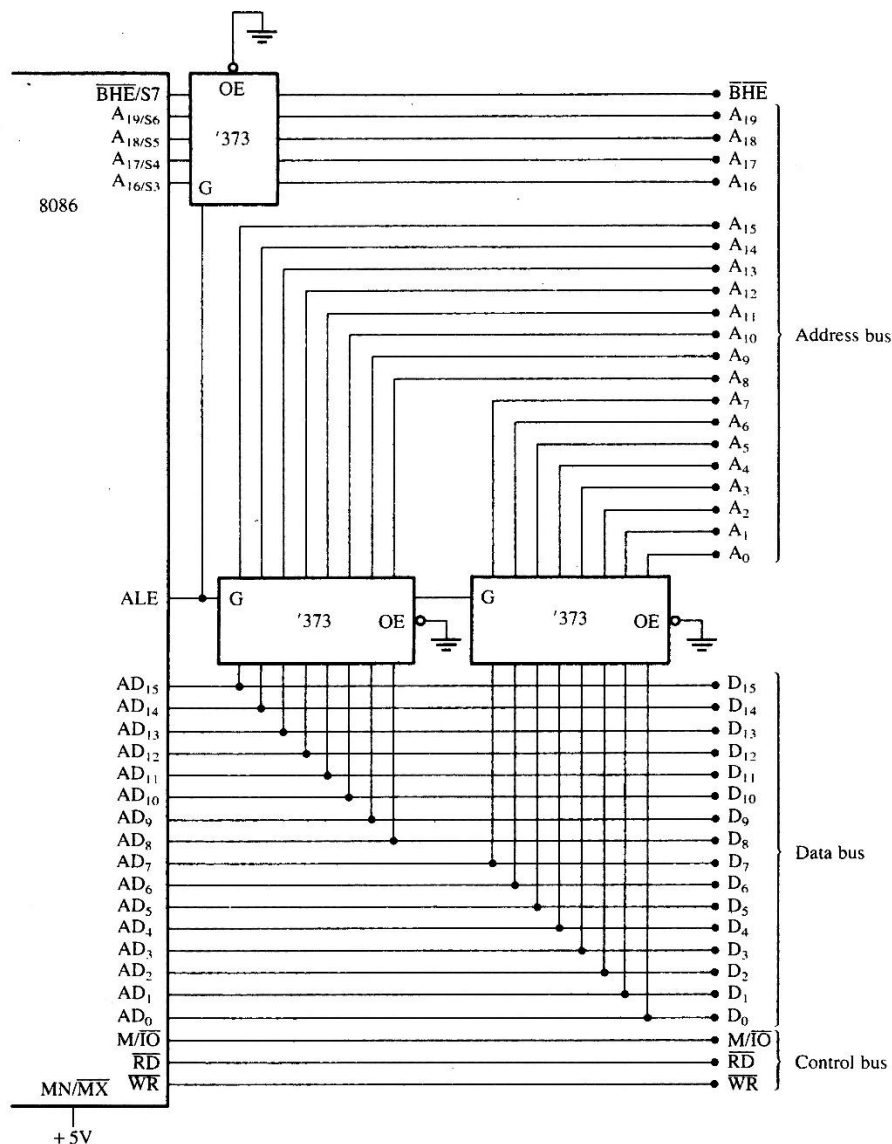
- The **address/data** bus of the 8086/8088 is **multiplexed** (shared) to reduce the number of pins required for the integrated circuit
 - the hardware designer must **extract** or **demultiplex** information from these pins
- Memory & I/O require the address remain valid and stable throughout a read/write cycle.
- If buses are **multiplexed**, the address changes at the memory and I/O, causing them to read or write data in the wrong locations

- All computer systems have **three buses**:
 - an **address bus** that provides memory and I/O with the memory address or the I/O port number
 - a **data bus** that transfers data between the microprocessor and the memory and I/O
 - a **control bus** that provides control signals to the memory and I/O
- These buses must be present in order to interface to memory and I/O.

Demultiplexing the 8086

- Fig 9–6 illustrates a demultiplexed 8086 with all three buses:
- **address** (A_{19} – A_0 and $\overline{BHE}$)
- **data** (D_{15} – D_0),
- **control** ($M/\overline{IO}$, $\overline{RD}$, and $\overline{WR}$)
- Here, the memory and I/O system see the 8086 as a device with:
 - a **20-bit address bus**; **16-bit data bus**
 - and a **three-line control bus**

Figure 9–6 The 8086 microprocessor shown with a demultiplexed address bus. This is the model used to build many 8086-based systems.



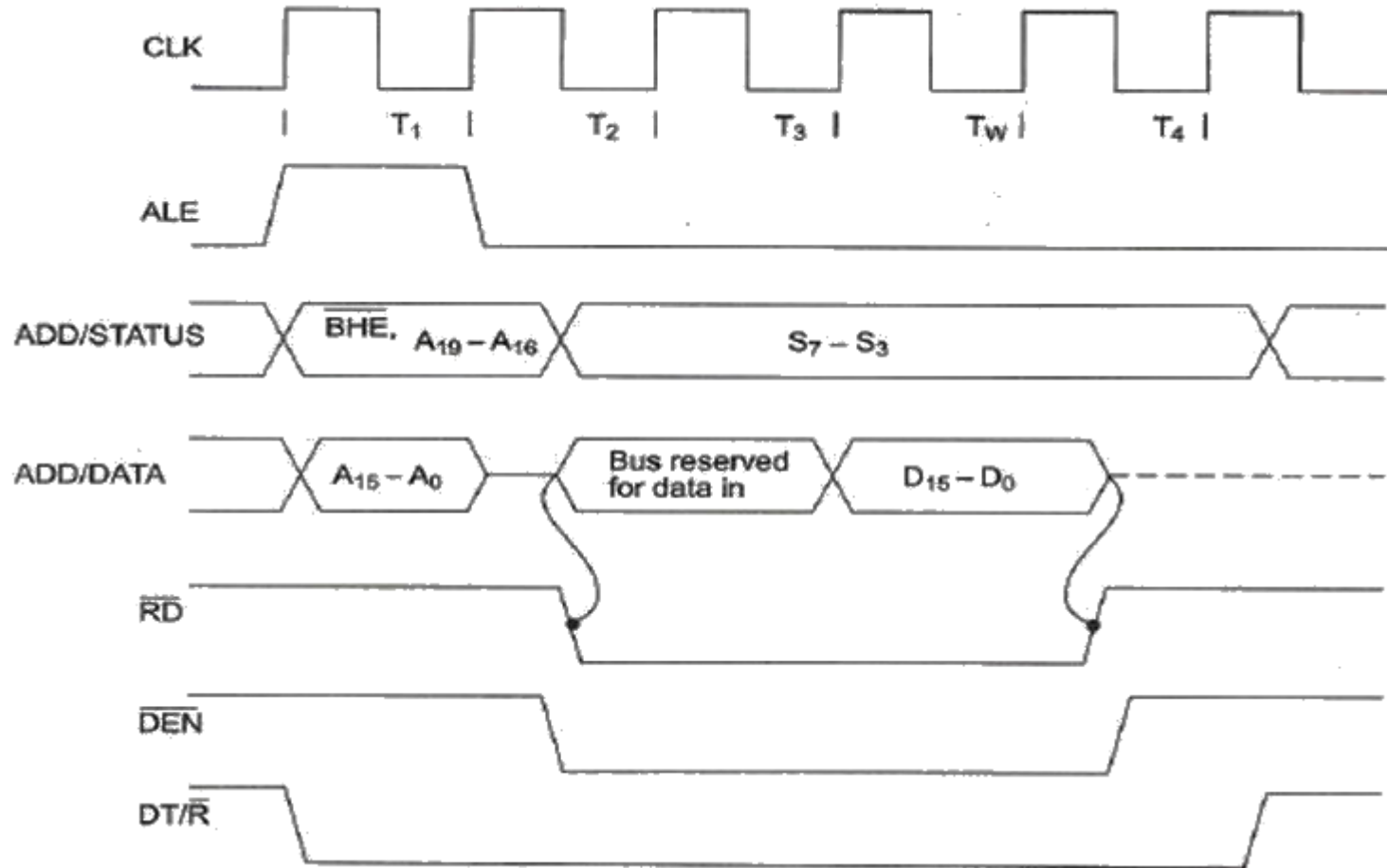
9-4 BUS TIMING

- It is essential to understand system **bus timing** before choosing memory or I/O devices for interfacing to 8086 or 8088 microprocessors.
- This section provides insight into operation of the bus signals and the **basic read/write timing** of the 8086/8088.

Basic Bus Operation

- The three buses of 8086 function the same way as any other microprocessor.
- If data are read from the memory the microprocessor:
 - **outputs** the memory address on the address bus
 - **issues** a **read** memory signal (RD)
 - and **accepts** the data via the data bus

Simplified 8086/8088 read bus cycle.



Basic Bus Operation

- If data are **written to memory** the processor:
 - **outputs** the memory address on the address bus
 - **outputs** the data to be written on the data bus
 - **issues** a write (WR) to memory
 - and IO/M = 1 for 8086